Mar 25, 2026

Nuclei Isolation with EZ Prep Kit

DOI
https://dx.doi.org/10.17504/protocols.io.36wgq7r8yvk5/v1

Le Zhang[1]

[1]Yale University

ASAP Collaborative Res...

N Nick C Buitrago-Pocasangre
Yale University

OPEN ACCESS

DOI: https://dx.doi.org/10.17504/protocols.io.36wgq7r8yvk5/v1

Protocol Citation: Le Zhang 2026. Nuclei Isolation with EZ Prep Kit. **protocols.io** https://dx.doi.org/10.17504/protocols.io.36wgq7r8yvk5/v1

License: This is an open access protocol distributed under the terms of the **Creative Commons Attribution License**, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited

Protocol status: Working
We use this protocol and it's working

Created: December 08, 2021

Last Modified: March 25, 2026

Protocol Integer ID: 55790

Keywords: Nuclei Isolation, EZ Prep Kit, ASAPCRN, nuclei isolation with ez prep kit, protocol details nuclei isolation, nuclei isolation, ez prep kit, nuclei, isolation, protocol detail, protocol

Funders Acknowledgements:
ASAP
Grant ID: ASAP-000529

Abstract

This protocol details nuclei Isolation with EZ Prep Kit.

Attachments

f3xvbkx77.pdf

92KB

Materials

- EZ Prep Kit

Troubleshooting

1 Prepare nuclei suspension buffer by adding 2 µL of RNase Inhibitor ([M] 40 U/µl) per mL of nuclei suspension buffer. Vortex briefly to mix.

2 Keeping everything On ice , add 1 mL EZ PREP buffer (Sigma, NUC101) and tissue piece to the small 2 mL glass tissue grinder. Grind 3 times just to mash the tissue at the bottom of the grinder, then homogenize 25 times with pestle A. Add an additional 1 mL EZ PREP buffer and homogenize 25 times with pestle B. Move slowly to avoid spilling buffer.

3 Use a pipette to transfer to 15 mL conical tube. Add 2 mL EZ PREP buffer to the tissue homogenizer to rinse and pipette into conical tube. Pipette gently to mix.

4 Incubate On ice for 00:05:00 . 5m

5 Centrifuge for 00:05:00 at 500 x g at 4 °C . 5m

6 Remove supernatant carefully with a pipette. Do not disturb the pallet and leave behind some buffer. Resuspend in 1 mL EZ PREP buffer, then add 3 mL EZ PREP buffer to wash.

7 Incubate 4 °C for 00:05:00 . 5m

8 Centrifuge for 00:05:00 at 500 x g at 4 °C . 5m

9 Remove supernatant carefully with a pipette. Resuspend in 1 mL nuclei suspension buffer, then add 3 mL nuclei suspension buffer. Pipette to mix.

Note

Do not disturb the pallet and leave behind some buffer.

10 Centrifuge for 00:05:00 at 500 x g and 4 °C . 5m

11 Remove supernatant carefully with a pipette. Resuspend in 500 µL nuclei suspension buffer.

12 In a new labeled Eppendorf tube, make a 1:1 dilution of the nuclei suspension and Trypan Blue and count nuclei with hemocytometer slide (C-Chip, DHCN015).

Note

Average count x 2 (dilution in Trypan Blue) x 10^4 /mL.

12.1 If there are visible clumps of nuclei when counting, re-pipette gently 20 times and then incubate On ice for 00:05:00 . Make new 1:1 dilution with Trypan Blue and then recount and check for clumps of nuclei. 5m